

Quality Control and Control Charts

Vaibhavi Z
Statistics with R

Introduction to Process Variation

- Every manufacturing process produces items with some variation.
- Even under strict control, variation cannot be completely eliminated.
- This inherent variation is called **chance variation**.

Types of Variation

1. Chance Variation

- Natural and unavoidable
- Inherent to the process

2. Assignable Variation

- Caused by specific factors
- Leads to poor quality

Causes of Assignable Variation

- Faulty machine settings
- Poor quality raw materials
- Incorrect software
- Human errors
- Other external factors

- A process is **in control** when only chance variation exists.
- A process is **out of control** when assignable variation occurs.
- Main goal: Determine whether the process is in or out of control.

Control Charts

- Control charts help monitor process stability.
- Based on:
 - Upper Control Limit (UCL)
 - Lower Control Limit (LCL)
- Data is divided into subgroups.

Subgroup Statistics

- Subgroup average (mean)
- Subgroup standard deviation
- These statistics are plotted on control charts.

Decision Rule:

- If value lies outside limits $\rightarrow$ Process is out of control

Types of Control Charts

For Measurable Characteristics:

- Mean ($\bar{X}$ -bar) chart
- Standard deviation chart

Assumption:

- Mean and variance are constant when process is in control

Attribute Control Charts

- Used when items are classified as:
 - Acceptable
 - Unacceptable
- Helps detect quality changes

Defect-Based Control Charts

- Used when items have multiple defects
- Number of defects is random
- Helps monitor defect rates

Advanced Control Charts

- Consider past subgroup values
- More sensitive to small changes

Types:

- Moving Average Chart
- Exponentially Weighted Moving Average (EWMA)
- Cumulative Sum (CUSUM) Chart

Conclusion

- Variation is unavoidable in manufacturing
- Control charts help detect abnormal variations
- Ensures quality and process stability

- Consider a manufacturing process producing items with measurable characteristics.
- When the process is **in control**:
 - Observations are independent
 - Normally distributed
 - Mean = μ , Variance = σ^2
- Process may go **out of control** due to special causes.

Objective

- Detect when the process goes out of control.
- Stop the process and identify the issue.
- Maintain quality and consistency.

Subgroup Formation

- Data is divided into subgroups of size n .
- Typical values: $n = 4, 5, 6$
- Subgroups ensure:
 - Uniform conditions within subgroup
 - Variation occurs between subgroups

Subgroup Averages

Let $X_1, X_2, \dots$ denote observations.

$$\bar{X}_1 = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$\bar{X}_2 = \frac{X_{n+1} + \dots + X_{2n}}{n}$$

$$\bar{X}_3 = \frac{X_{2n+1} + \dots + X_{3n}}{n}$$

When process is in control:

$$E(\bar{X}_i) = \mu$$

$$Var(\bar{X}_i) = \frac{\sigma^2}{n}$$

$$Z = \frac{\bar{X}_i - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Control Limits Derivation

For standard normal variable:

$$P(-3 < Z < 3) = 0.9973$$

Thus:

$$-3 < \frac{\sqrt{n}(\bar{X}_i - \mu)}{\sigma} < 3$$

$$\mu - \frac{3\sigma}{\sqrt{n}} < \bar{X}_i < \mu + \frac{3\sigma}{\sqrt{n}}$$

Upper Control Limit (UCL):

$$UCL = \mu + \frac{3\sigma}{\sqrt{n}}$$

Lower Control Limit (LCL):

$$LCL = \mu - \frac{3\sigma}{\sqrt{n}}$$

Center Line (CL):

$$CL = \mu$$

X-bar Control Chart

- Plot subgroup averages $\bar{X}_i$
- Draw three lines:
 - UCL
 - CL
 - LCL
- Monitor process behavior over time

- If $\bar{X}_i$ lies within limits $\rightarrow$ Process is in control
- If $\bar{X}_i$ lies outside limits $\rightarrow$ Process is out of control

Action:

- Stop process
- Identify cause
- Take corrective measures

Problem Statement

A manufacturer produces steel shafts having diameters that should be normally distributed with mean 3 mm and standard deviation 0.1 mm. Successive samples of four shafts have yielded the following sample averages in millimeters. Sample averages (in mm):

3.01, 2.97, 3.12, 2.99, 3.03, 3.02, 3.10, 3.14, 3.09, 3.20

What conclusion should be drawn?

Step 1: Control Limits

Formula:

$$UCL = \mu + \frac{3\sigma}{\sqrt{n}}, \quad LCL = \mu - \frac{3\sigma}{\sqrt{n}}$$

Substitute values:

$$UCL = 3 + \frac{3(0.1)}{\sqrt{4}} = 3 + 0.15 = 3.15$$

$$LCL = 3 - \frac{3(0.1)}{\sqrt{4}} = 3 - 0.15 = 2.85$$

Step 2: Control Limits Summary

- Upper Control Limit (UCL) = 3.15
- Lower Control Limit (LCL) = 2.85
- Center Line (CL) = 3

Step 3: Compare Sample Means

Sample	Mean
1	3.01
2	2.97
3	3.12
4	2.99
5	3.03
6	3.02
7	3.10
8	3.14
9	3.09
10	3.20

Step 4: Analysis

- Control limits: $2.85 \leq \bar{X} \leq 3.15$
- All samples lie within limits except:
 - Sample 10: $\bar{X} = 3.20$

Conclusion

- Sample 10 exceeds the upper control limit.
- Indicates presence of assignable cause.
- Process is **out of control**.

Conclusion:

- Mean diameter may have shifted from 3 mm.
- Investigation and corrective action required.

Problem Statement

A manufacturing company produces electronic components. The probability that a component is defective is 3%. Each day, a sample of 150 components is inspected.

- (a) Suggest a suitable control chart.
- (b) Calculate the control limits.
- (c) If 10 defectives are found, comment on process control.

(a) Suitable Control Chart

Since we are counting the number of defective items in a fixed sample size:

- Use **np-chart**

(b) Control Limits

Given:

$$p = 0.03, \quad n = 150$$

Mean:

$$np = 150 \times 0.03 = 4.5$$

Standard deviation:

$$\sigma = \sqrt{np(1-p)} = \sqrt{150 \times 0.03 \times 0.97} = \sqrt{4.365} \approx 2.09$$

Upper Control Limit:

$$UCL = np + 3\sigma = 4.5 + 3(2.09) = 10.77$$

Lower Control Limit:

$$LCL = np - 3\sigma = 4.5 - 6.27 = -1.77 \approx 0$$

(c) Conclusion

Observed defectives = 10

$$0 < 10 < 10.77$$

- The value lies within control limits
- **Process is under control**